75. Assertion (A) : The main constituent of the Liquefied Petroleum Gas is methane.
Reason (R) : Methane can be used directly for burning in homes and factories where it can be supplied through pipelines.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is correct but R is wrong.
(d) A is wrong but R is correct.

Ans. (d) : The assertion is incorrect as Liquefied Petroleum Gas (LPG) consists mainly of propane, propylene, butane, and butylene in various mixtures. These gases are present at normal temperature and pressure. It does not contain methane. Methane cannot be used directly for burning in homes and factories.

76. Consider the following statements:
1. Liquefied Natural Gas (LNG) in liquefied under extremely cold temperatures and high pressure to facilitate storage or transportation in specially designed vessels.
2. First LNG terminal in India was built in Hassan.
3. Natural Gas Liquids (NGL) are separated from LPG and these include ethane, propane, butane and natural gasoline.
Which of the statements given above is/are correct?
(a) 1 only (b) 1 and 3
(c) 2 and 3 (d) 1, 2 and 3

Ans. (b) : First LNG terminal in India was built in Dahej, Gujarat and another terminal at Kochi, Kerala. Hence statement 2 is not correct. Rest of the statements is correct.

77. Which one of the following is the correct sequence of the given substances in the decreasing order of their densities?
(a) Steel > Mercury > Gold
(b) Gold > Mercury > Steel
(c) Steel > Gold > Mercury
(d) Gold > Steel > Mercury

Ans. (b) : Gold > Mercury > Steel
Density of gold is 19.30 g/cm3.
The density of ultra pure liquid mercury is 13.534 g/cm3.
The density of steel is 7.80 g/cm3.

78. Virtual water trade is being looked at by experts as a solution to the world's water crisis. What does virtual water (VW) imply.
(a) Volume of heavy water required to replace ordinary water.
(b) Volume of water required to produce a commodity or service.
(c) Volume of water saved by using rain-water harvesting.
(d) Volume of water utilized by an effective flood control.

Ans. (b) : Virtual water trade (also known as trade in embedded or embodied water) refers to the hidden flow of water if food or other commodities are traded from one place to another. It refers to the sum of the water use in the various steps of the production chain.

BIOLOGY

79. Pneumoconiosis affects the workers who work mainly in:
(a) Tanneries
(b) Coal mining industry
(c) Distilleries
(d) Glass industry

Ans. (b) : Pneumoconiosis, a lung disease, mostly affects workers who work in the mining and construction sector and deals with soil, silica, coal dust and asbestos. The disease includes asbestosis, silicosis and coal workers' pneumoconiosis.

80. Assertion (A) : All the proteins in our food are digested in small intestine only.
Reason (R) : The protein-digesting enzymes from pancreas are released into small intestine.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is correct but R is wrong.
(d) A is wrong but R is correct.

Ans. (d) : The digestion of protein begins in stomach. The enzyme pepsin acts on protein and brings about their breakdown in the stomach. The protein-digesting enzymes from pancreas are released into small intestine.

81. Assertion (A) : Ameoba reproduces by fission.
Reason (R) : All unicellular organisms reproduced by asexual methods.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is correct but R is wrong.
(d) A is wrong but R is correct.

Ans. (a) : Assertion is correct and reason is the correct explanation of assertion. Amoeba is a unicellular animal. It reproduce by the process of asexual reproduction. Fission is one of the processes of asexual reproduction, where amoeba divides in half to produce off spring.

82. Assertion (A) : The person with diabetes insipidus feels thirsty.
Reason (R) : A person with diabetes insipidus suffers from excess secretion of vasopressin.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is correct but R is wrong.
(d) A is wrong but R is correct.

Ans. (a) : Diabetes Insipidus (DI) is a condition characterized by excessive thirst and excretion of large amounts of severely diluted urine. Diabetes mellitus is a group of metabolic diseases in which a person has high blood sugar, due to two causes either the pancreas does not produce enough insulin, or because cells do not respond to the insulin that is produced. The most common type in humans is the neurological form which involves a deficiency of arginine vasopressin or antidiuretic hormone.

83. Which of the following diseases of milching animals are infectious?
1. Foot and Mouth disease 2. Anthrax
3. Black Quarter 4. Cowpox
Select the correct using the code given below:
(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1 and 4 (d) 1, 2, 3 and 4

Ans. (c) : Black Quarter is a disease which is found in horse.
Anthrax is a serious infectious disease caused by gram-positive, rod-shaped bacteria known as Bacillus anthracis. Anthrax can be found naturally in soil and commonly affects domestic and wild animals around the world.

84. Consider the following statements:
1. Dengue is a protozoan disease transmitted by mosquitoes.
2. Retro-orbital pain is not a symptom of Dengue.
3. Skin rash and bleeding from nose and gums are some of the symptoms of Dengue.
Which of the statements given above is/are correct?
(a) 1 and 2 (b) 3 only
(c) 2 only (d) 1 and 3

Ans. (c) : Dengue is a viral disease transmitted through mosquito. Fever, headache, pain in joints, back muscles and eyeballs, skin rash for a few days, bleeding from nose and gums are some of the symptoms of the dengue fever.

ENVIRONMENT AND ECOLOGY

85. Consider the following statements:
1. The forest cover in India constitutes around 20% of its geographical area. Out of the total forest cover, dense forest constitutes around 40%
2. The National Forestry Action Programme aims at Bringing one-third of the area of India under tree/forest cover.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) : Out of the total forest cover, dense forest constitutes around 2.54%; very dense forest and 8.77% are moderately dense forest. National Forestry Action Programme aims a long term plan to achieve the target of 33% forest cover.

CURRENT AFFAIRS

86. Consider the following statements:
1. Kyoto Protocol came into force in the year 2005.
2. Kyoto Protocol deals primarily with the depletion of the Ozone layer.
3. Methane as a greenhouse gas is more harmful than carbon dioxide.
Which of the statements given above is/are correct?
(a) 1 and 2 (b) 1 and 3
(c) 1 only (d) 3 only

Ans. (b) : The Kyoto Protocol is an international agreement that aimed to reduce carbon dioxide (CO_2) emissions and the presence of greenhouse gases (GHG) in the atmosphere. The essential tenet of the Kyoto Protocol was that industrialized nations needed to lessen the amount of their Green House Gases emissions.
The Kyoto Protocol is an international agreement linked to the United Nations Framework Convention on Climate Change, which commits its parties by setting internationally binding greenhouse gases emission reduction targets. Therefore statement 2 is incorrect. The Kyoto Protocol was adopted in Kyoto, Japan on 11 December, 1997 and entered into force on 16 February, 2005. Hence statement 1 is correct. Methane traps up to 100 times more heat in the atmosphere than carbon dioxide, but the fact is that methane also leaves the atmosphere within a decade. This makes methane a short-lived but intense climate changer. So methane warms the planet rapidly, but it dissipates from the atmosphere more quickly than carbon dioxide. Therefore statement 3 is also correct.

87. Who among the following is not a recipient of the Bharat Ratna Award?
(a) Ustad Bismillah Khan
(b) Satyajit Ray
(c) Lata Mangeshkar
(d) Raj Kapoor

Ans. (d) : Raj Kapoor is not a recipient of Bharat Ratna award. Lata Mangeshkar 2001, Bismillah Khan 2001, and Satyajit Ray 1992 are the holders of Bharat Ratna Award.

88. Who among the following is the President of the FICCI?
(a) Sunil Mittal
(b) Brijmohan Lal Munjal
(c) Onkar S. Kunwar
(d) Vivek Burman